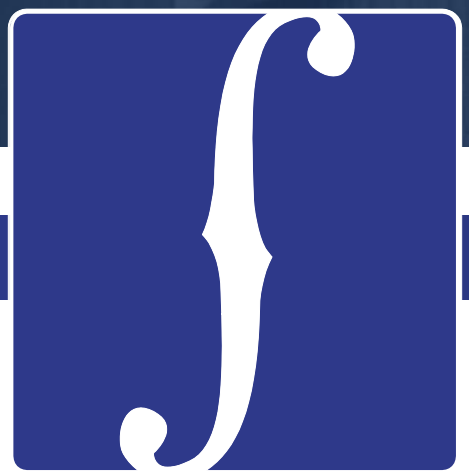




CELLO

Mortgage Insurance

April 2016

Mortgage lenders protect themselves against default-related losses by requiring **mortgage insurance** for loans with LTVs about 80%. There are two main providers of mortgage insurance in the U.S.: the government (FHA/VA/RHS) and private mortgage insurance companies.

The most prevalent form of mortgage insurance is **flow insurance** – insurance available on a loan-by-loan basis at origination.

Other forms of insurance include **bulk insurance**, **pool insurance**, and **reinsurance**.

Mortgage Insurance (MI)

Generally required for any single-family residential mortgage loan with an LTV ratio greater than 80%. By charter, Fannie Mae and Freddie Mac are not allowed to purchase residential mortgages with LTVs over 80% without credit enhancement. Bank regulations require credit support for mortgages on 1-4 family residential properties with LTVs equal to greater than 90%. MI is generally not available for loans with LTVs > 100%.

Key Providers

- **Federal Housing Administration (FHA):** Government-backed program. Insures loans originated to borrowers who can afford to make only minimal downpayments and/or have issues with credit.
- **Veteran's Authority (VA):** Government-backed program. Insures loans originated to homeowners who served or are serving in the Armed Forces.
- **Private Mortgage Insurers (PMIs):** Privately-owned companies that provide mortgage insurance in various forms including on a loan-by-loan basis, in bulk, and at the pool-level. PMIs also provide reinsurance.

There are multiple delivery channels for MI. The government is only involved in the flow channel, the private industry operates in all channels.

Primary Mortgage Insurance

- **Flow:** Loans insured one at a time; premiums based on filed rate sheets
- **Bulk:** Many loans insured at the same time. Each loan is priced separately with final rates set through a bidding process.

Pool Mortgage Insurance

- Many loans insured through a single pool policy. Rates determined through a bidding process. All loans have the same premium rate.
- Commonly have aggregate deductibles and stop loss thresholds

Reinsurance

- The primary insurer transfers (**cedes**) a portion of the periodic insurance premium to the reinsurer in return for the reinsurer's commitment to share losses. In some cases, the reinsurer also pays an upfront fee to the primary insurer.
- Losses are shared on a pro-rata (**quota share**) or above a certain threshold (**excess of loss**).
- Most private mortgage reinsurance is written by **captive** reinsurers affiliated with the lender.
- Motivation for reinsurance is regulatory capital requirements.
- Primary insurer remains liable for all coverage if the reinsurer fails to pay.

Key terms:

NOD (Notice of Delinquency): Servicer must notify insurer that an insured loan is in arrears. A NOD does not necessarily equate to a claim.

Cure: A NOD that is rectified without a mortgage insurance claim.

NIW (New insurance written): Balance on new policies written

Risk: Coverage \$ on insured loans

	Comments
Capital Intensive	Need \$3 of capital to write \$1 of premium. Ratios have increased recently.
Indefinite Policy Terms	Policy term is unknown at the time the loan is originated and averages 5-6 years (the typical WAL for 30-year loans).
Contagion	At times of economic distress, risks of mortgage default become highly correlated as many borrowers are simultaneously impacted.
Small claim sizes	A typical claim size is \$50K and a \$200K claim is rare. Losses are driven by frequency.
Loss emergence is not uniform	The likelihood of defaulting can vary significantly over the life of the policy.
Short-tailed reserving	Most delinquent loans are resolved within 12 to 18 months (Delinquency resolutions were much slower in the Housing Crisis of 2007-2009 and its aftermath).

Given the statutory requirements of the GSEs and the housing policy objectives of the FHA and VA insurance programs, MI plays an integral role in the U.S. Housing Finance System.

In particular, changes in the availability and level of MI affect homeowner prepayment rates and thus impact the pricing of MBS.

Key Connections Between MI and Prepayments

- **Availability of MI:** MI availability plays a key role in High LTV (LTVs > 80%) lending. The entry or exit of MI providers can have a profound impact on the ability of High LTV borrowers to refinance their loans.
- **Changes in MI Underwriting Standards:** This is an extension to the first point. Opening up/closing the underwriting box, changing the relative pricing of collateral characteristics in the underwriting matrix, and changing the overall level of MI premiums significantly influences which subsets of High LTV borrowers can refinance.
- **Cross-Sector Refinances:** For LTVs between 90-97%, the government and private MI programs can be regarded as competitors for the same pool of High LTV borrowers. Changes in pricing by one program can thus affect the refinancing behavior of borrowers in both programs.

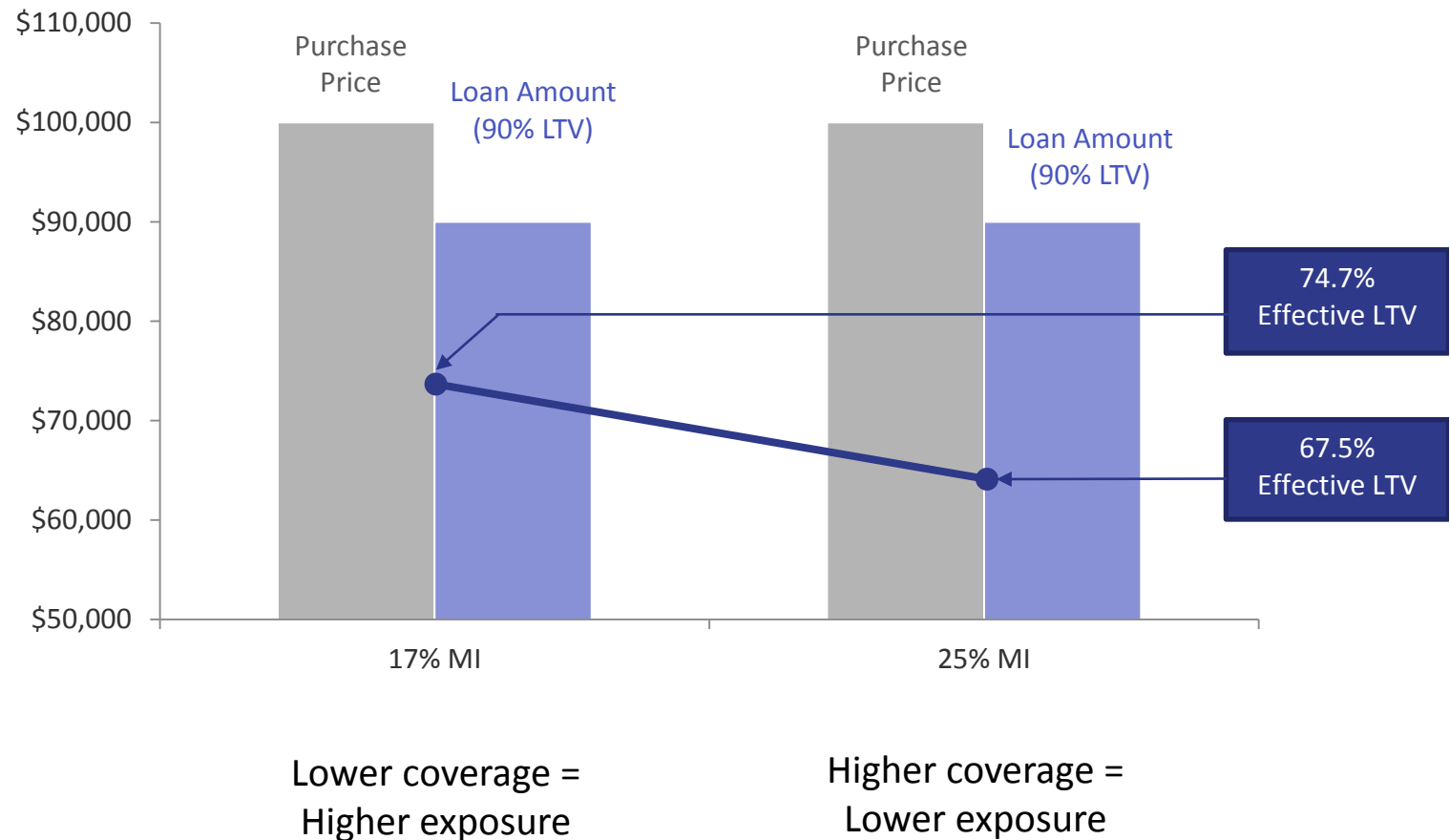
Factor	Private Mortgage Insurance	Government Programs
Minimum down payment	Typically 3%.	3.5% on FHA-insured mortgages. In most cases VA does not require a down payment.
Loan coverage level	Varies	FHA insures the entire loan balance; VA guarantees a percentage of the loan.
Limit on size of the mortgage insured or guaranteed	None	Beginning in 2009, the loan limit for FHA-insured mortgages for one-unit properties is 115 percent of the local area median home price, as determined by HUD, with a floor of 65 percent of the conforming loan limit (or \$271,050) and a ceiling equal to 150 percent of the conforming limit . Limits vary by geographic region and for 2 - 4 unit properties. The VA's loan limits are the same as conforming loan limits.
Up-front mortgage insurance premium	Generally not required	1.75% UFMIP Required by FHA; VA requires an up-front funding fee.
Monthly premium	Premiums vary based on the size of the down payment, type of mortgage, and amount of insurance coverage.	FHA borrowers pay an annual insurance premium that starts at 0.85% percent of the loan balance and declines over time. There is no insurance premium for VA mortgages.
Cancellation of insurance	Can usually be canceled when the homeowner acquires 20 percent equity in the home. Under Federal law, MI must be cancelled automatically when the borrower has paid the loan down to 78 percent of the original home value.	For FHA mortgages with terms greater than 15 years, the annual mortgage insurance cancels when the LTV ratio reaches 78 percent, provided the borrower has paid the annual mortgage insurance premiums for at least 5 years and that the loan was closed after December 31, 2000 and had a case number assignment before June 3 rd , 2013.
Deductibility of borrower-paid mortgage insurance	Yes, up to a certain income level, through the 2016 tax year.	Yes, up to a certain income level, through the 2016 tax year.
Income limits	None	No maximum income limit on FHA/VA loans.
Interest rate	Market driven	Market driven
Regulation	State regulated. Regulation extends to reserves for losses, capital, etc.	FHA and VA mortgage programs are administered by agencies of the U.S. government.

A PMI covers only a specified portion of the loss. This percentage is called the **coverage** percentage and is typically between 12% to 35% of the **gross claim amount**.

The gross claim amount consists of the outstanding principal, accrued interest, and various expenses incurred during the foreclosure process.

PMI coverage lowers the LTV of the loan from the perspective of the holder of credit risk on the loan.

Effective LTV =
 $1 - \text{Equity} - (\text{Coverage} * \text{OLTV})$



Most originators use the loss coverage scheme developed by Fannie Mae and Freddie Mac.

Coverage	UNDERWRITTEN LTV RANGE		EFFECTIVE LTV RANGE	
35%	>= 95.01	<= 100	61.8>=	65%
30%	90.01	95	58.5	61.8
25%	85.01	90	55.3	58.5
12%	80	85	52.0	55.3

The MI premium is expressed as an annual percentage of the loan balance.

MI premiums may be incorporated into the borrower's monthly payment or paid separately.

Rates cannot change over the life of the policy and the policy cannot be cancelled by the insurer except for non-payment of premiums.

Fixed Rate 30-YEAR ■ NON-REFUNDABLE

For loans with level payments for the first 5 years

BASE LTV	COVERAGE	LOAN'S REPRESENTATIVE CREDIT SCORE			
		760+	720-759	680-719	620-679
97% to 95.01%	35%	1.05%	1.10%	1.31%	1.48%
	30	.90	.94	1.14	1.31
	25	.73	.77	.96	1.11
	18	.57	.60	.80	.94
95% to 90.01%	35	.63	.71	1.01	1.32
	30	.54	.62	.89	1.15
	25	.50	.57	.79	1.03
	18	.45	.51	.69	.80
	16	.43	.49	.67	.74
90% to 85.01%	30	.49	.54	.64	.83
	25	.39	.44	.57	.71
	17	.33	.38	.44	.56
	12	.29	.34	.39	.47
85% & UNDER	25	.37	.39	.43	.64
	17	.30	.32	.38	.53
	12	.23	.27	.33	.39
	6	.21	.25	.29	.33

STANDARD AGENCY
COVERAGE REQUIREMENTS

for loan terms > 20 years

for loan terms ≤ 20 years

PREMIUM ADJUSTMENTS	LOAN'S REPRESENTATIVE CREDIT SCORE			
	760+	720-759	680-719	620-679
Rate-and-Term Refinance	.05%	.10%	.15%	.30%
Cash-out Refinance	.18	.20	.25	.50
Second Homes	.12	.14	.20	.35
Loan Amounts > \$417,000 ¹	.20	.25	.40	.60
Refundable Monthly Premium	.01	.01	.02	.03
Declining Renewals	.02	.03	.04	.05
Employee Relocation Loans ²	-.02	-.04	-.07	-.10
≤ 25-Year Amortization Term ²	-.03	-.05	-.08	-.11
Manufactured Housing	.18	.20	.30	.50
Investment Property	.34	.38	.50	.75

¹For properties in Alaska and Hawaii, Loan Amounts > \$625,500.

²Rates cannot be reduced below .15%.

If a borrower defaults on an insured loan and the lender forecloses and takes title to the property, there will be an **insurance claim**.

The MI usually has the option to pay the entire loan balance and expenses to the lender and take title to the property or to let the lender retain the title and pay the lesser of:

- Actual losses plus expenses
- The coverage amount.

Statutory disbursements = statutory expenses related to the foreclosure process such as sale publications, title costs, filing fees, deed recordings, or sheriff's expenses for conducting the sale.

Original Loan Balance, LTV, Coverage	\$90,000, 90% LTV, 25% Coverage
Principal Balance Due at Default	\$88,915
Accumulated Interest	7,850
Subtotal	\$96,765
Expenses	
Attorney's Fees	\$1,000
Property Taxes	1,140
Hazard Insurance	710
Property Preservation	350
Statutory Disbursements	\$1,500
Subtotal	\$101,465
Less Escrow Balance	(\$240)
Total Possible Claim	\$101,225
Times Coverage Purchase (25%)	× .25
Total Optional Claim Payment	\$25,306

Over time, PMI premiums for higher credit borrowers (as captured by the FICO score) have decreased, whereas those for lower credit borrowers have increased.

These changes are partly in response to the new capital requirements for the PMIs (PMIERS) rolled out by the GSEs.

May 2014									
LTV Range	Coverage	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
95.01 97	35%	1.10	1.20	1.20	1.46	1.46	1.78	1.78	1.78
90.01 95	30%	0.59	0.72	0.72	1.04	1.04	1.45	1.45	1.45
85.01 90	25%	0.44	0.54	0.54	0.72	0.72	1.01	1.01	1.01
0.01 85	12%	0.28	0.37	0.37	0.48	0.48	0.69	0.69	0.69
April 2016									
LTV Range	Coverage	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
95.01 97	35%	0.55	0.75	0.95	1.15	1.40	1.90	2.05	2.25
90.01 95	30%	0.41	0.59	0.73	0.87	1.08	1.42	1.50	1.61
85.01 90	25%	0.30	0.41	0.50	0.60	0.73	1.00	1.05	1.10
0.01 85	12%	0.19	0.20	0.23	0.27	0.32	0.41	0.43	0.45
Difference: May 2014 - April 2016									
LTV Range	Coverage	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
95.01 97	35%	0.55	0.45	0.25	0.31	0.06	-0.12	-0.27	-0.47
90.01 95	30%	0.18	0.13	-0.01	0.17	-0.04	0.03	-0.05	-0.16
85.01 90	25%	0.14	0.13	0.04	0.12	-0.01	0.01	-0.04	-0.09
0.01 85	12%	0.09	0.17	0.14	0.21	0.16	0.28	0.26	0.24

The PMI and FHA programs are mostly competitive in the 90-97% LTV range. The PMI programs are more targeted towards higher FICO scores.

Assumptions:

- 1% LLPA = 25bps in yield
- 1% UFMIP = 8bps in yield (because the UFMIP can be rolled into the loan balance)

LTV=97%, PMI Coverage = 35%		Credit Scores						
	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
FHA								
Upfront MI Premium	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%
Annual Premium	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%
Conforming								
GSE LLPA	0.75%	0.75%	1.00%	1.50%	1.50%	2.25%	2.75%	3.50%
PMI Annual Premium	0.55%	0.75%	0.95%	1.15%	1.40%	1.90%	2.05%	2.25%
FHA Advantage	-0.17%	0.03%	0.29%	0.62%	0.87%	1.55%	1.83%	2.22%

LTV=95%, PMI Coverage = 30%		Credit Scores						
	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
FHA								
Upfront MI Premium	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%
Annual Premium	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%
Conforming								
GSE LLPA	0.25%	0.25%	0.50%	1.00%	1.25%	2.25%	2.75%	3.25%
PMI Annual Premium	0.41%	0.59%	0.73%	0.87%	1.08%	1.42%	1.50%	1.61%
FHA Advantage	-0.44%	-0.26%	-0.05%	0.21%	0.48%	1.07%	1.28%	1.51%

LTV=90%, PMI Coverage = 25%		Credit Scores						
	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
FHA								
Upfront MI Premium	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%
Annual Premium	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%
Conforming								
GSE LLPA	0.25%	0.25%	0.50%	1.00%	1.25%	2.25%	2.75%	3.25%
PMI Annual Premium	0.30%	0.41%	0.50%	0.60%	0.73%	1.00%	1.05%	1.10%
FHA Advantage	-0.55%	-0.44%	-0.29%	-0.06%	0.13%	0.65%	0.83%	1.00%

LTV=85%, PMI Coverage = 12%		Credit Scores						
	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
FHA								
Upfront MI Premium	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%
Annual Premium	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%	0.85%
Conforming								
GSE LLPA	0.25%	0.25%	0.50%	1.00%	1.25%	2.25%	2.75%	3.25%
PMI Annual Premium	0.19%	0.20%	0.23%	0.27%	0.32%	0.41%	0.43%	0.45%
FHA Advantage	-0.66%	-0.65%	-0.55%	-0.39%	-0.28%	0.06%	0.21%	0.35%

Assumptions

Loan Amount	\$250,000
Loan Purpose	Rate-term Refinance
Occupancy	Owner-occupied
Property Type	Single-family

Base Rate

Conforming	3.71%
FHA	3.63%

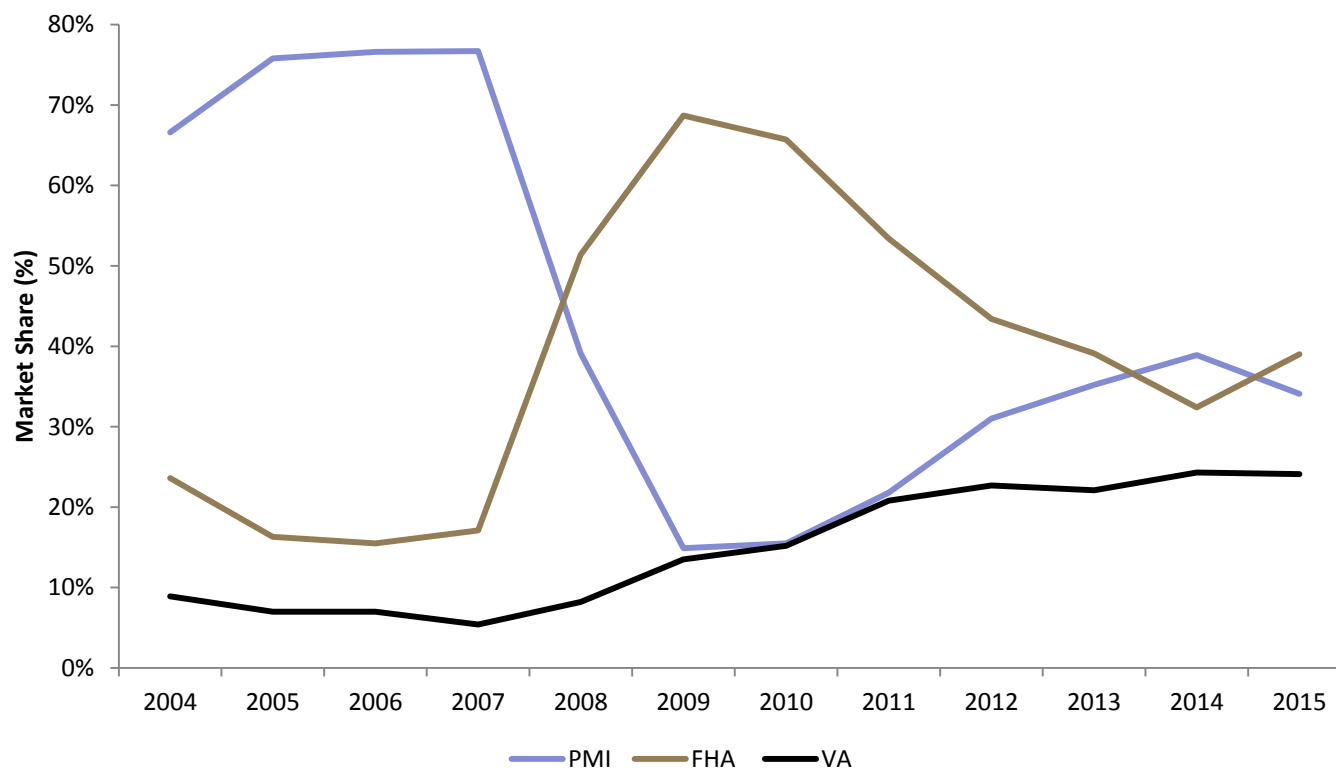
The PMI share of Primary MI was very high prior to the Housing Crisis of 2007-2009 as a significant number of borrowers migrated away from the FHA program to private-label sectors such as Alt-A and Subprime.

Over 2007-2009, the collapse of the housing markets and the ensuing issues that private mortgage insurers markedly increased FHA's market share.

Since 2009, there has been a gradual increase in PMI share and a steady decrease in FHA share as private mortgage insurers were gradually restored to health.

In 2015, FHA's declining market share was reversed by a cut in its mortgage insurance premiums.

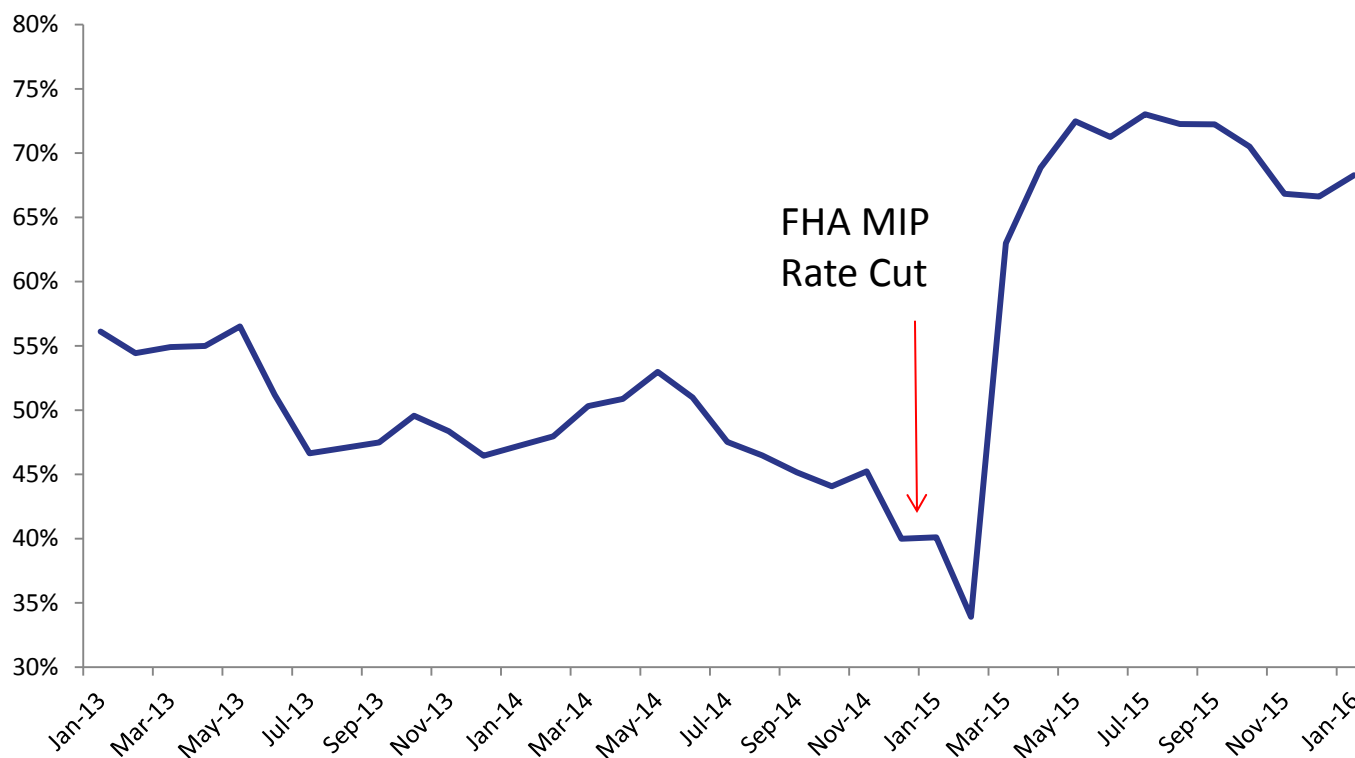
NEW PRIMARY MORTGAGE INSURANCE MARKET SHARE (2004-2015)



The PMIs and the FHA compete for the same borrowers in the 90-97% LTV range.

Changes in the pricing of one program can thus have an impact on the prepayment rates of both programs.

FHA SHARE OF HIGH LTV REFINANCINGS (January 2013 – January 2016)



Conventional High LTV loans: All newly originated refinanced loans with LTVs >80% across all FNMA/FHLMC 30-year pool types (CL, CQ, U6, U9 etc.)

FHA: All newly originated 30-year FHA Refinance loans.

There are currently 7 active providers of MI in the U.S.

Although there are significant gaps in the availability of historical MI pricing across private mortgage insurers, it is relatively straightforward to get a sense for current pricing.

Corporation	2015 Primary Market Share*	URL	Comments
United Guaranty	24%	https://www.ugcorp.com/	Fill in web form to obtain quotes
Mortgage Guaranty Insurance Corp	20	https://www.mgic.com/rates/rate-cards.html	Resource for historical MI levels
Radian Guaranty Inc.	19	http://www.radian.biz/page?name=RatesAndGuidelinesNew	"BPMI Prime Rates"
Genworth Financial	15	https://mortgageinsurance.genworth.com/RatesAndGuidelines/RateCards.aspx	"Monthly Premium MI"
Essent Guaranty	12	https://www.essent.us/rates-guidelines	"Borrower-Paid Mortgage Insurance Monthly"
National MI	6	http://www.nationalmi.com/products-rates/	
Arch Mortgage Insurance	5	https://mi.archcapgroup.com/Rates/Rate-Sheets	

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